

Adaptive Discrepancy Masked Distillation for remote sensing object detection

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ARTICLE INFO

Keywords:

Object detection
Knowledge distillation
Remote sensing images

ABSTRACT

Knowledge distillation (KD) has become a promising technique for obtaining a performant student detector in remote sensing images by inheriting the knowledge from a heavy teacher detector. Unfortunately, not every pixel contributes (even detrimental) equally to the final KD performance. To dispel this problem, the existing methods usually derived a distillation mask to stress the valuable regions during KD. In this paper, we put forth Adaptive Discrepancy Masked Distillation (ADMD), a novel KD framework to explicitly localize the beneficial pixels. Our approach stems from the observation that the feature discrepancy between the teacher and student is the essential reason for their performance gap. With this regard, we make use of the feature discrepancy to determine which location causes the student to lag behind the teacher and then regulate the student to assign higher learning priority to them. Furthermore, we empirically observe that the discrepancy masked distillation leads to loss vanishing in later KD stages. To combat this issue, we introduce a simple yet practical weight-increasing module, in which the magnitude of KD loss is adaptively adjusted to ensure KD steadily contributes to student optimization. Comprehensive experiments on DIOR and DOTA across various dense detectors show that our ADMD consistently harvests remarkable performance gains, particularly under a prolonged distillation schedule, and exhibits superiority over state-of-the-art counterparts. Code and trained checkpoints will be made available at <https://github.com/swift1988>.

1. Introduction

Object detection for remote sensing image (RSI) has experienced a flourish, thanks to the tremendous development of deep convolutional neural networks (Simonyan and Zisserman, 2014; He et al., 2016; Lai et al., 2023). To further promote the detection performance, however, modern detectors (Ren et al., 2015; Lin et al., 2017b; Tian et al., 2019; Xie et al., 2021; Yao et al., 2022; Yan et al., 2024; Xie et al., 2024) are always equipped with heavy backbones, which unavoidably introduce extra storage and computational usage, rendering them to be too complicated to deploy in resource-constraint platforms, such as mobile devices, aerial surveillance, etc. To break through this issue, a large body of techniques are proposed to derive a lightweight model with competitive performance, including pruning (He et al., 2017), lightweight model design (Sandler et al., 2018), and knowledge distillation (KD) (Hinton et al., 2015; Adriana et al., 2015; Zhu et al., 2021; Li et al., 2023a). In this article, we focus on KD, which aims to transmit the knowledge of an over-parameters teacher model to a compact student model.

Knowledge distillation was originally devised for the image classification task and popularized by Hinton et al. (2015). Recently, there

has been a wave of adapting KD to object detection where the feature imitation (Adriana et al., 2015) is commonly adopted (Wang et al., 2019; Yang et al., 2022a; Huang et al., 2023). However, recklessly imitating all pixels with the same priority is less efficient since not every pixel equally contributes (even detrimental (Wang et al., 2019; Zhixing et al., 2021)) to the final KD performance (Huang et al., 2023). To handle this issue, recent studies dedicated to exploring valuable areas for distillation and attained certain performance (Li et al., 2023b; Yang et al., 2022d; Zhang and Ma, 2021; Yang et al., 2022a,c; Huang et al., 2023; Zhixing et al., 2021; Yang et al., 2023). For example, assuming the foreground is more important for object detection, Yang et al. (2022c), Wang et al. (2019), Yang et al. (2022d), Guo et al. (2021) stressed the regions near the ground-truth. Similarly, Yang et al. (2022a), Zhang and Ma (2021) leveraged attention maps to identify the foreground areas, and Li et al. (2023b) extracted an instance prototype to locate the instance-relevant pixels. Some works attempted to discern favorable features by measuring teacher classification scores (Zhixing et al., 2021; Yang et al., 2024), learning receptive tokens (Huang et al., 2023), or using the gradient information (Lan and Tian, 2024).

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<https://doi.org/10.1016/j.isprsjprs.2025.02.006>

Received 12 October 2024; Received in revised form 6 February 2025; Accepted 7 February 2025

Available online 26 February 2025

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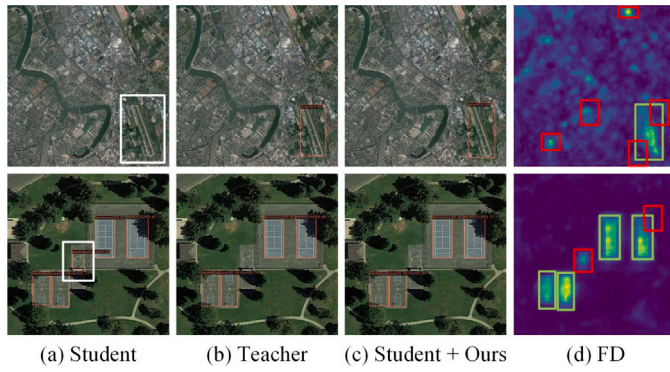


Fig. 1. Visualization of the detection results and the feature discrepancy (FD) between the student (FCOS-ResNet18) and the teacher (FCOS-ResNet101). The white boxes in (a) mark the undistilled student's wrong predictions. In (d), the red boxes indicate that FD retrieves profitable information from the background and removes misleading information from the foreground. The green boxes in (d) are the ground-truth. More examples of spatial FD for each FPN layer can be found in Fig. 6. Best viewed in color with zoom-in.

Nevertheless, the previous methods suffer from the following drawbacks, leaving how to completely choose beneficial features for distillation remains an unsolved problem. (i) **Unclear key distillation regions.** The foreground-oriented distillation masks overlook the profitable patches in the background (Guo et al., 2021; Li et al., 2023b) and introduce misleading features in the foreground (Zhixing et al., 2021). This problem is especially aggravated under RSIs, which exhibit more complex data characteristics between the foreground and background (Li et al., 2020a; Xia et al., 2018; Cheng et al., 2021b; Li et al., 2023b). In essence, for a dense prediction task, it is ill-posed to determine the major KD areas by simply distinguishing foreground and background. (ii) **Single-scale mask for multi-scale imitation.** The prior works typically construct a single-scale mask (Guo et al., 2021; Yang et al., 2022c; Li et al., 2023b; Wang et al., 2019). However, current dense detectors consist of feature pyramid networks (FPN) (Lin et al., 2017a) to build multi-scale features, which have different regions of interest. Thus, hastily using the single-scale distillation mask to guide the multi-scale imitation may bring redundant information and fall into a local optimum. (iii) **Ignoring channel-wise mask.** Many efforts (Wang et al., 2019; Guo et al., 2021; Li et al., 2017, 2023b; Zhixing et al., 2021; Yang et al., 2022a; Zhang and Ma, 2021; Yang et al., 2022c,d; Huang et al., 2023) are limited to only derive the spatial-wise distillation mask. However, different channels reflect the discriminative information of different categories (Hu et al., 2018; Cheng et al., 2023). Therefore, assigning identical distillation priority for each channel constrains the effects of dense KD.

To tackle the above problems, we put forth a new perspective on generating the distillation mask to further facilitate the detection KD under RSIs. Specifically, as illustrated in Fig. 1, the feature discrepancy (FD) strongly correlates with the performance gap between detectors. This observation gives us two proper inspirations, which have not been explored for distilling detectors yet. Firstly, we conjecture that if the student could perfectly minimize the feature discrepancy with the teacher, *regardless of foreground or background*, the student's performance would be significantly promoted. Secondly, FD reflects how much each pixel lags behind the teacher; namely, FD could tell us how much effort the student should give to each pixel during the distillation. Taking the two inspirations together, we are prompted to calculate the FD along the training process and straightforwardly adopt the FD as the mask to instruct the distillation. By doing so, we explicitly down-weight the learning priority assigned to the easy locations and enlarge the contribution from the samples with significant discrepancies, consequently facilitating the student to catch up with the teacher. In practice, our distillation is executed at both the feature and

logit levels to reach better results.

Our discrepancy-oriented distillation mask offers the following advantages and effectively circumvents the limitations outlined above. (i) Our FD does not rely on the priors of bounding boxes, which allows us to collect concealed informative areas from the background and filter out meaningless pixels from the foreground, as shown in Fig. 1. (ii) FD is independently generated for each FPN level with the corresponding FPN feature, enabling our method to be more suitable for modern detectors. (iii) Beyond the spatial-wise FD, channel-wise FD is instinctively exploited to find important distillation channels.

In addition, we also empirically identify a bottleneck raised in the discrepancy-oriented mask. Specifically, due to the FD becoming smaller and smaller as the distilling progresses, the students' distilling effort will be damped, especially in the later training stage (See Fig. 3(b)). As a result, the overall distillation effects are suppressed. To mitigate this issue, we propose an adaptive weight module, in which a monotonically increasing function is introduced to adjust the overall KD loss's weight explicitly. Whilst simple, the distillation results evidence that this technique can bring us practical benefits.

Instantiated in the two designs proposed above, we establish a simple yet efficient detection KD framework, dubbed Adaptive Discrepancy Masked Distillation (ADMD). To evaluate ADMD, we conduct comprehensive experiments on two standard RSI datasets: DIOR (Li et al., 2020a) and DOTA (Xia et al., 2018) over several dense detectors with various backbones. ADMD achieves promising distillation results and consistently exceeds the state-of-the-art KD methods. The major contributions of this article are summarized below:

- A discrepancy-oriented distillation mask is proposed to aid the student in mimicking the teacher's outputs, both the feature and logit levels.
- A loss adjustment module is introduced to warrant the distillation loss consistently contributing to the student's optimization.
- Building upon our two practical designs, we create an efficient detection KD framework, in which each module could provide new insight for future research.

2. Related works

2.1. Object detection

Object detection is regarded as one of the most challenging tasks in computer vision, aiming at classifying the categories and regressing the coordinates of the instances in a given image. Modern detection frameworks can be divided into one-stage (Lin et al., 2017b) and two-stage detectors (Ren et al., 2015; Xie et al., 2021, 2023; Sun et al., 2024). Two-stage detectors, represented by Faster RCNN (Ren et al., 2015), develop the region proposal network (RPN) to produce a set of candidate proposals and process each proposal to obtain final results. While one-stage detectors, e.g., FCOS (Tian et al., 2019), directly predict the category and localization of targets, achieving a better trade-off between speed and accuracy. Besides, another criterion can group detectors into anchor-based (Lin et al., 2017b) and anchor-free (Tian et al., 2019; Duan et al., 2019) ones, according to whether performing detection with the preset sliding windows, i.e., anchor boxes.

The complex characteristics of RSIs pose new challenges to object detection (Cheng et al., 2016, 2022a,c; Yao et al., 2022; Cheng et al., 2021b; Sun et al., 2023; Yao et al., 2024b,a; Cheng et al., 2018) in handling large ratios, orientations, and scales variation of instances (Li et al., 2020a). To tackle these obstacles, specialized object detection methods for RSIs have been proposed. For instance, Xu et al. (2021) found that the feature misalignment limits the detection performance, then proposed a feature aligned single-shot detector. Cheng et al. (2021a) proposed a feature enhancement and context encoding modules to promote inter-class separation. Cheng et al. (2022b) presented an anchor-free scheme to generate high-quality and oriented proposals. To circumvent the boundary discontinuity problem, Yao et al. (2022) put forward a new bounding box representation for oriented object detection.

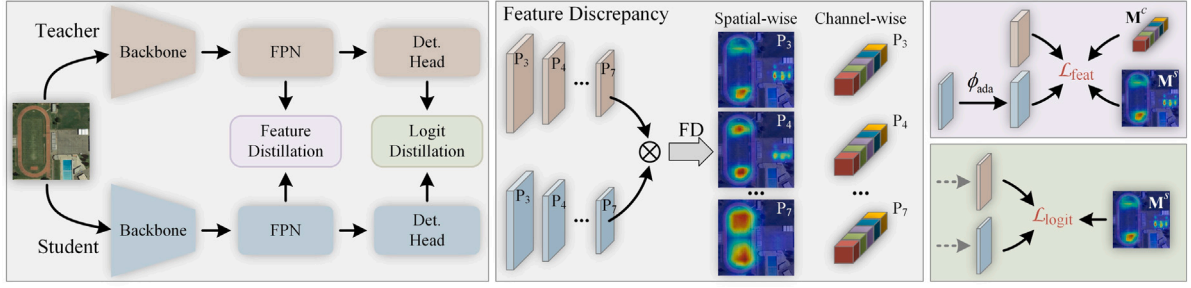


Fig. 2. Details of the proposed method. Along the training process, we independently generate the spatial-wise and channel-wise distillation masks with the corresponding FPN features of the teacher (frozen) and the student (tuned). Then for each FPN layer and the subsequent classification head, the masked feature-based KD and logit-based KD are performed, respectively. The $\otimes$ is a pair-wise distance function (e.g., cosine distance) to measure the discrepancy of features (FD), and ϕ_{ada} is the adaptation layer (e.g., 1×1 Conv) to align the channel of the student to the teacher.

2.2. Knowledge distillation on object detection

Knowledge distillation (KD) is a model compression technique to learn a lightweight but performant student model by inheriting knowledge from a cumbersome teacher (pre-trained) network. Regarding what is to be distilled, three representative patterns of knowledge are developed, including output probability (Hinton et al., 2015), intermediate feature (Adriana et al., 2015), and relational information among the features (Park et al., 2019). Due to its effectiveness and utility, KD has been successfully advanced in various computer vision tasks, including image classification (Hinton et al., 2015), semantic segmentation (Yang et al., 2022e), and object detection (Chen et al., 2017; Li et al., 2023b). Instead of further developing KD, Li et al. (2024) presented a framework that searches for optimal distillers for different student–teacher pairs, improving the applicability of existing KD methods to new scenarios and models. This work focuses on applying KD for the object detection task.

Distilling detectors can be traced back to Chen et al. (2017), in which the knowledge is directly transferred between the teacher and student. Wang et al. (2019) first pointed out that simply performing full feature transmission would hurt the student and put forward a fine-grained feature-based distillation method. In the spirit of Wang et al. (2019), the primary concerns in the detection KD boil down to realizing more precise foreground excavation (Zhang and Ma, 2021; Zhixing et al., 2021; Guo et al., 2021; Yang et al., 2022c,b,a; Zheng et al., 2022; Li et al., 2023b). For example, Zhang and Ma (2021) and Yang et al. (2022a) leveraged the attention mechanisms to gather the foreground pixels and emphasized them during distillation. Zhixing et al. (2021), Yang et al. (2024, 2023), Wang et al. (2024) relied on the classification scores to collect informative features that can promote the detectability of small models. Yang et al. (2022c) proposed an adaptive multi-scale features selection strategy to guide the student to learn the core target regions only. Li et al. (2023b) presented an instance-aware distillation method in which a parameter-free module is developed to produce a fine-grained imitation mask. However, these methods heavily rely on the ground-truth or anchor, and it is hard to identify which pixels are really conducive to transferring knowledge. Our method, in contrast, bypasses these dilemmas and chooses the discrepancy of features between the student and teacher as the criterion to construct the imitation mask dynamically. Guo et al. (2021) also advocated the benefit of the background. However, their work treated all the background as the informative regions, while our method could only retrieve the profitable information from the background. Besides, we empirically discover that the discrepancy masking would suppress the KD effects in the later training stage. To alleviate this problem, we proposed an adaptive weight adjustment strategy, which brings noticeable performance gains, especially under a prolonged KD schedule.

2.3. Weight adjustment in knowledge distillation

Determining the overall weight of distillation loss is critical for training students (Hu et al., 2024; Deng and Zhang, 2021; Kim et al., 2021; Deng et al., 2022). Specifically, Hu et al. (2024) exploited both the correctness of the teacher and student to decide *sample-wise* weight. In self-KD, Deng and Zhang (2021), Deng et al. (2022), Kim et al. (2021) gradually increases the KD weight as the student itself becomes more confident about their outputs during training. In this paper, with distinct observation and motivation, we adjust the distillation weight to guarantee the KD loss term constantly contributes to the student's learning.

3. Methodology

The overall framework of the proposed approach can be found in Fig. 2. Our method follows two main principles: (1) finding the pixels with large feature discrepancy between the teacher and student; and (2) tilting the distillation towards large discrepancy locations. Concretely, we leverage the feature discrepancy to generate the mask (Section 3.2) and then execute discrepancy masked distillation on both the FPN feature (Section 3.3) and the classification head (Section 3.4). Besides, we adaptively adjust the loss weight to ensure the KD effects in the whole distillation course (Section 3.5).

3.1. Preliminaries

3.1.1. Notations

Here, we first list the main notations used in this section to ease the presentation. We tag the feature of each FPN as $\mathbf{F} \in \mathbb{R}^{H \times W \times K}$ and the logit (generated in the classification head) as $\mathbf{z} \in \mathbb{R}^{H \times W \times C}$, where H , W , K , and C indicate the height, width, channel and the number of categories, respectively. We discriminate $\mathbf{F}$ and $\mathbf{z}$ by the superscripts S and T for the students and the teachers.

3.1.2. KD for object detection

For distilling detectors, the typical manner is to instruct the student to mimic the teacher's outputs pixel-wisely and the KD objective is abstracted as:

$$\mathcal{L}_{\text{KD}} = D(f^S, f^T) \quad (1)$$

where f is the model's output and it can be the feature $\mathbf{F}$ or logit $\mathbf{z}$ in detection KD. D is the task-specific distance function. However, equally emulating each pixel would weaken the detection KD performance due to the severe imbalance of object scales and the partition of foreground and background (Wang et al., 2019; Huang et al., 2023). One promising solution for this issue is to derive a distillation mask $\mathbf{M}$ to highlight the worthy locations, and Eq. (1) is reformulated as:

$$\mathcal{L}_{\text{KD}} = \mathbf{M} \cdot D(f^S, f^T) \quad (2)$$

In this paper, we propose a new perspective to generate the distillation mask, without any priors of bounding boxes and hyper-parameters. Next, without loss of generality, we elaborate our method based on one-stage dense detectors under a single FPN level.

3.2. Discrepancy between the student and teacher

Given the FPN features $\mathbf{F}^S$ and $\mathbf{F}^T$, we quantify the spatial and channel FD between the students and teachers with *cosine distance*, obtaining $\mathbf{M}^s \in \mathbb{R}^{H \times W}$ and $\mathbf{M}^c \in \mathbb{R}^K$:

$$\mathbf{M}_{i,j}^s = \cos(\mathbf{F}_{i,j}^S, \mathbf{F}_{i,j}^T) \quad (3)$$

$$\mathbf{M}_k^c = \cos(\delta(\mathbf{F}_k^S), \delta(\mathbf{F}_k^T)) \quad (4)$$

where (i, j) and k denote the spatial and channel indices, respectively. The superscripts s and c stand for the spatial and channel. $\delta(\cdot)$ is used to flatten the features. The *cosine distance* defined as $\cos(\mathbf{u}, \mathbf{v}) = 1 - \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} \in [0, 2]$. The $\cos(\cdot)$ is our default choice, and we include a limited ablation of the distance function for the curious reader in Section 4.3.5.

Through the $\mathbf{M}^s$ and $\mathbf{M}^c$, we can easily identify the points that the students lag behind the teacher. In practice, we directly opt for the in-hand $\mathbf{M}^s$ and $\mathbf{M}^c$ as the distillation mask, and it works surprisingly well. Additionally, modern detectors always equip FPN to handle the instances of various sizes. In this regard, it is conducive to each FPN level possessing its own distillation mask. Thus, with Eqs. (3) and (4), we calculate the $\mathbf{M}^s$ and $\mathbf{M}^c$ with the corresponding FPN feature for KD on the FPN layer and the subsequent classification head.

3.3. Feature distillation

Feature-based distillation has been justified as an effective method of delivering the localization knowledge captured by the teacher's features. Following the prior works, we choose the FPN feature for distillation. Nevertheless, empirical results indicate that directly employing feature distillation gets trivial improvements for the dense detection task. To alleviate this issue, we propose a discrepancy-oriented mask to sample the most valuable areas to transfer. Formally, with the generated distillation mask $\mathbf{M}^s$ and $\mathbf{M}^c$, the objective of feature imitation can be generally written as follows:

$$\mathcal{L}_{\text{feat}} = \sum_{i=1}^H \sum_{j=1}^W \mathbf{M}_{i,j}^s \cdot \sum_{k=1}^K \mathbf{M}_k^c \cdot \mathcal{L}_{\text{mse}}(\phi_{\text{ada}}(\mathbf{F}_{i,j,k}^S), \mathbf{F}_{i,j,k}^T) \quad (5)$$

where $\phi_{\text{ada}}(\cdot)$ is an adaptation layer to align¹ the feature's shape between the students and teachers, e.g., 1×1 convolutional layer used in our paper, and $\mathcal{L}_{\text{mse}}(\cdot)$ is the commonly used pixel-wise Mean Square Error.

3.4. Logit distillation

Similar to feature distillation, executing the distillation on the whole logit map may weaken the final KD performance. In particular, excessive easy samples (e.g., background) would disturb the student training and lead to inferior detectors (Zhixing et al., 2021; Yang et al., 2023). Therefore, it is also necessary to retrieve informative locations for logit distillation. In this paper, we directly opt for the FD as the distillation mask and the reasons are two folds. (i) When a pixel is perfectly matched in the feature space, we could regard it as an easy one, and otherwise. That is to say, with FD, it is reasonable and convenient to discrete the easiness and hardness of positions during

¹ According to the empirical analysis in Chen et al. (2017), even the features of teachers and students having the same number of channels, adding ϕ_{ada} is also recommended in distilling detectors.

Table 1

Comparison with state-of-the-arts on DIOR (Li et al., 2020a) and DOTA (Xia et al., 2018). 1x, 20e, and 2x indicate the models are trained with 12, 20, and 24 epochs, respectively.

Dataset	DIOR			DOTA		
Method	FCOS	GFocal	ATSS	FCOS	GFocal	ATSS
Teacher	68.78	65.65	69.18	62.54	60.43	63.50
Student	61.98	57.44	62.57	56.58	56.82	57.64
+KD (Hinton et al., 2015)	63.93	59.91	65.84	56.58	56.82	57.64
+DeFeat Guo et al. (2021)	66.16	58.49	66.33	59.23	58.41	58.39
+FKD (Zhang and Ma, 2021)	66.28	58.21	65.04	58.10	58.02	58.42
+CWD (Shu et al., 2021)	65.83	61.69	65.21	58.31	56.85	58.20
+FRS (Zhixing et al., 2021)	65.27	61.15	65.78	59.14	58.29	58.54
+ARSD (Yang et al., 2022c)	64.97	61.22	65.63	58.74	57.61	58.63
+S ³ MKM (Yang et al., 2022d)	65.72	61.51	65.32	59.15	57.87	58.55
+FGD (Yang et al., 2022a)	65.88	61.23	66.45	59.38	58.06	58.62
+MGD (Yang et al., 2022b)	66.28	61.90	67.16	59.73	58.22	58.38
+PKD (Cao et al., 2022)	65.20	61.42	64.63	56.60	57.32	58.34
+InsDist (Li et al., 2023b)	66.72	60.96	66.09	59.89	58.28	58.17
+DRKD (Ni et al., 2023)	66.03	60.44	66.53	59.27	58.07	58.49
+MasKD (Huang et al., 2023)	66.29	61.76	66.61	59.41	58.04	58.74
+BCKD (Yang et al., 2023)	66.32	62.06	66.72	59.12	58.30	59.06
+GKD (Lan and Tian, 2024)	66.17	61.95	65.91	59.02	57.86	58.89
+TWA (Yang et al., 2024)	66.29	61.86	65.87	59.32	57.92	59.31
+CrossKD (Wang et al., 2024)	66.54	62.43	66.44	59.49	58.11	59.66
1 x	67.05	62.87	67.42	60.06	58.51	60.28
+ADMD (Ours) 20e	68.04	63.16	67.73	60.61	58.72	60.52
2 x	68.14	63.35	68.01	61.43	58.94	60.77

distillation. (ii) The behaviors of the classification head vary in detectors, e.g., per-proposal in the anchor-based method (Ren et al., 2015) and per-location in the anchor-free one (Tian et al., 2019). Therefore, the in-hand FD allows our KD framework more versatile in different scenarios. To this end, we formally express the logit matching process as follows:

$$\mathcal{L}_{\text{logit}} = \sum_{i=1}^H \sum_{j=1}^W \mathbf{M}_{i,j}^s \cdot \mathcal{L}_{\text{bce}}(\mathbf{z}_{i,j}^S, \mathbf{z}_{i,j}^T) \quad (6)$$

where $\mathcal{L}_{\text{bce}}(\cdot)$ represents the Binary Cross-Entropy loss. Note that although the Kullback–Leibler (KL) divergence is commonly used in logit KD, we empirically find $\mathcal{L}_{\text{bce}}(\cdot)$ leads to better distillation for detectors. A similar observation was also made in Yang et al. (2023), Zhixing et al. (2021).

3.5. Adaptive weight adjustment

In the prior methods, the mask is usually unchanged so the KD weight is set to a fixed value. However, in the proposed ADMD, the element in FD (our distillation mask) is dynamically changed and becomes smaller and smaller along the distillation process, especially in the later distillation stage. This undesired phenomenon will lower the distilling effort, consequently diminishing the effect of KD. Thus, in ADMD, to ensure the KD steadily contributes to the training, it instinctively increases the overall KD weight. In light of this idea, we gradually update the weights every epoch as follows:

$$\lambda^{(n)} = \lambda \cdot (\log_{\mu} n + 1) \quad (7)$$

where $\lambda^{(n)}$ is the distillation weight in the n th epoch, and $\mu > 1$ is a tunable hyper-parameter. λ is initialized and fixed before training (See Eq. (8)).

3.6. Overall loss function

To summarize, the overall loss is model-agnostic and formulated as follows:

$$\mathcal{L}_{\text{overall}} = \mathcal{L}_{\text{det}} + \lambda_1 \mathcal{L}_{\text{feat}} + \lambda_2 \mathcal{L}_{\text{logit}} \quad (8)$$

where $\mathcal{L}_{\text{det}}$ is the task loss for detectors, λ_1 , λ_2 are the initial weight coefficient and updated with Eq. (7). We empirically set λ_1 to 2×10^{-2} and λ_2 to 1.0 by default.

4. Experiment

4.1. Setups

4.1.1. Datasets

We evaluate the effectiveness and robustness of our method on two large-scale remote sensing object detection datasets, namely DIOR (Li et al., 2020a) and DOTA (Xia et al., 2018). (i) DIOR contains 23,463 images with the shape of 800×800 across 20 categories. We use 11,725 images for training and 11,738 images for evaluation. (ii) DOTA is divided into 15 semantic classes. The images are cropped into fixed 1024×1024 with 200 pixels overlap, resulting in 15,749, 5,297, and 10,833 images for training, validation, and testing.

4.1.2. Implementation details

All experiments are performed under mmdetection (Chen et al., 2019) framework with the default configure files. Following Yang et al. (2022c,d, 2023), Zheng et al. (2022), unless other specified, we adopt the dense one-stage detectors with ResNet101 and ResNet18 as the backbones for the teachers and students, and train the models for 12 epochs ($1 \times \text{schedule}$) with a batch size of 16. The momentum and weight decay are set as 0.9 and 0.0001, respectively. All models are initialized with ImageNet pre-trained weights. We report the results in terms of average precision (AP) using intersection over union (IoU) at 0.5.

4.2. Comparison with state-of-the-arts

We report the quantitative results for DIOR (Li et al., 2020a) and DOTA (Xia et al., 2018) in Tables 1. Several state-of-the-art methods are picked for comparison (Hinton et al., 2015; Guo et al., 2021; Zhang and Ma, 2021; Yang et al., 2022c,d,a; Zhixing et al., 2021; Yang et al., 2022b; Li et al., 2023b; Huang et al., 2023; Yang et al., 2023; Cao et al., 2022; Yang et al., 2024; Lan and Tian, 2024; Wang et al., 2024). Those results illustrate that significant performance improvements are achieved over the undistilled students, demonstrating the effectiveness of our method. For DIOR (Li et al., 2020a), ADMD obtains 5.07%, 5.43% and 4.85% mAP improvements for FCOS (Tian et al., 2019), GFocal (Li et al., 2020b) and ATSS (Zhang et al., 2020), respectively. For DOTA (Xia et al., 2018), the enhancement is 3.48%, 1.69%, and 2.64%. Besides, ADMD consistently outperforms the prior KD approaches with a clear margin, demonstrating that our distillation mask effectively collects the beneficial pixels and stresses them during KD. Furthermore, using a longer training schedule, e.g., 20 epochs, aids the student in obtaining further performance enhancement.

4.3. Ablation studies

Unless other stated, we carry out the ablative studies of the proposed method with FCOS-ResNet18 as the student and FCOS-ResNet101 as the teacher over DIOR (Li et al., 2020a).

4.3.1. Effect of each component

We first investigate the contributions of each loss term on DIOR (Li et al., 2020a) and DOTA (Xia et al., 2018). The corresponding results are summarized in Table 2. We observe that both feature distillation and logit distillation could bring impressive detection improvements, e.g., on DIOR, feature KD and logit KD obtain notably 4.89% and 4.16% absolute mAP gains, respectively. Especially, we note that only applying feature distillation already surpasses the recent approaches, validating its effectiveness in stimulating the student. Furthermore, feature and logit distillation can complement each other and we achieve the best performance by combining them.

Table 2

Ablation study of each component.

$\mathcal{L}_{\text{det}}$	$\mathcal{L}_{\text{feat}}$	$\mathcal{L}_{\text{logit}}$	DIOR	DOTA
✓			61.98	56.58
✓	✓		66.87	59.89
✓		✓	66.14	59.17
✓	✓	✓	67.05	60.06

Table 3

Ablation study of the spatial and channel distillation mask.

Spatial	Channel	FCOS		GFocal		ATSS	
		mAP	Δ	mAP	Δ	mAP	Δ
✓		61.98	–	57.44	–	62.57	–
✓		66.38	4.40	61.12	3.68	67.21	4.64
	✓	66.26	4.28	61.14	3.70	66.95	4.38
✓	✓	66.87	4.89	62.68	5.24	67.29	4.72

4.3.2. Spatial FD and channel FD

In this article, we use both the spatial and channel-wise FD to instruct the feature imitation. Here, we explore the effects of each distillation mask. As listed in Table 3, each mask is capable of facilitating the students. Particularly, only distilling with the channel mask, which is generally overlooked in prior publications (Zhixing et al., 2021; Li et al., 2023b; Yang et al., 2024; Huang et al., 2023; Guo et al., 2021; Wang et al., 2019), could also bring 4.28%, 3.70%, 4.38% mAP gains for FCOS (Tian et al., 2019), GFocal (Li et al., 2020b), and ATSS (Zhang et al., 2020), respectively. These results justify that exploring the channel-wise distillation mask is crucial for feature imitation and the effectiveness of the proposed method. Our ADMD achieves the best performance by synergizing these two masks.

4.3.3. Weight adjustment analysis

The adaptive weight adjustment (AWA) module, another critical design in our KD framework, aims to preserve the KD effect in the later learning stage. Here, we validate its impact and study the proposed method's sensitivity to the hyper-parameter μ in Eq. (7). Fig. 3 plots the corresponding weight growing curve, distillation losses, and performance trend along the optimization procedure. Table 4 summarizes the distillation results for several values of μ . The following observations could be made. (i) As shown in Fig. 3(b), using the fixed weight (i.e., without AWA) causes the distillation loss to drop quickly. The main reason for this phenomenon is that our feature discrepancy (FD) mask is dynamically calculated during the distillation course, therefore the exact value of the FD will become smaller and smaller as the KD processes. (ii) Smaller FD would down-weight the overall effort in distilling knowledge from the teacher, consequently inhibiting the effectiveness of KD and leading to an unsatisfactory result eventually. In particular, as the learning curve shown in Fig. 3(c), the KD performance of directly distilling with FD starts to lag behind the case of applying our AWA strategy in the later KD stages. (iii) As shown in Table 4, The proposed AWA module certainly leads to a better performance, regardless of the choice of $\mu \in [1.5, 5]$. These experimental results verify the effectiveness of guaranteeing the student consistently puts effort into matching the teacher. (iv) We found that a moderate $\mu = 2$ yields the best results in our experiments. We ascribe this phenomenon to the fact that a lower $\mu = 1.5$ produces overwhelming distillation loss, disturbing the training process and negatively affecting the student's learning. In comparison, a higher $\mu = 5$ causes the modulating factor to be too smooth to achieve the expected weight increase, thus resulting in an inferior solution as well. Note that only feature distillation is applied here for simplicity, with $\lambda_1 = 2 \times 10^{-2}$. To reduce the labor of tuning hyper-parameters, in practice, we adopt the same μ for adjusting λ_1 and λ_2 , and it works surprisingly well. (v) The central tenet of AWA is to increase the KD weight along the distillation process, and we perform it every epoch by default. Intuitively, the iteration-level adjustment could lead to better KD results, and we leave it for our future work.

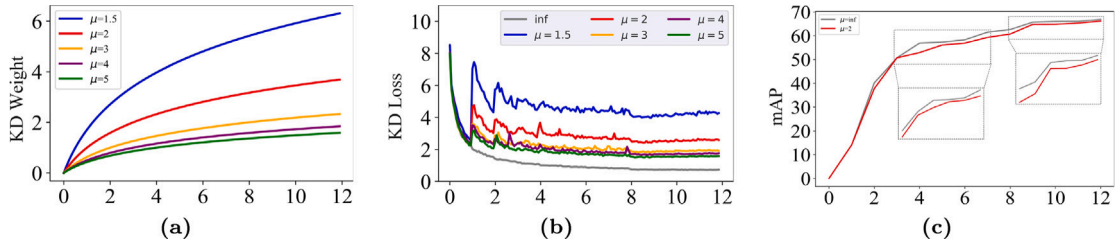


Fig. 3. (a) plots the weight growing trend by varying μ , (b) shows the corresponding distillation losses, and (c) is the performance curve. Legend ‘inf’ in (b) indicates the loss weight is fixed. The X-axis refers to the epoch number.

Table 4

Comparisons of different values of μ on DIOR with FCOS. ‘inf’ indicates the loss weight is fixed. ResNet101 and ResNet18 are the backbones for the teacher and the student, respectively.

μ	inf	1.5	2	3	4	5
mAP	66.18	66.56	66.87	66.73	66.58	66.31

Table 5

Hyper-parameter sensitivity of λ with FCOS on DIOR. ResNet101 and ResNet18 are the backbones for the teacher and student.

λ_1	μ	mAP	λ_2	μ	mAP
0.1	1.1	65.75	0.1	1.1	65.45
0.5	1.2	66.62	0.5	1.1	65.98
1.0	2.0	66.83	1.0	2.0	66.14
1.5	2.0	66.85	1.5	2.0	66.12
2.0	2.0	66.87	2.0	2.0	66.00
2.5	2.0	66.82	2.5	3.0	65.91
3.0	4.0	66.51	3.0	4.0	66.02

Table 6

Impact of The (a) FD measurement, and the (b) adaptation layer on DIOR. Here, only feature matching is applied. ResNet101 and ResNet18 are the backbones for the teacher and the student, respectively.

(a) Type of FD		(b) Type of ϕ_{ada}	
Type	mAP	Type	mAP
L1-Norm	65.85	1×1 conv	66.87
KL-divergence	64.79	1×1 conv + bn	66.77
Euclidean	66.01	3×3 conv	66.57
Cosine	66.87	3×3 conv + bn	66.82

4.3.4. Sensitivity of λ

The hyper-parameter of λ is used to balance the overall distillation loss. We conduct the sensitivity of study on DIOR (Li et al., 2020a) in Table 5. For a fair comparison, we search the optimal μ for each case. We find that the λ is correlated with the μ , as they determine the volume of distillation loss together. Generally, a higher λ picks a higher μ . Overall, however, the benefit of changing λ is insensitive, except for a very low of λ , and keeps fairly robust to exact λ/μ combinations.

4.3.5. Impact of FD measurement

In Section 3.2, the *cosine distance* is our default selection to measure the discrepancy of the features. Such a metric can also be quantified by other commonly used pairwise functions, e.g., euclidean distance, KL-divergence, and L1-norm distance, and Table 6(a) reports the performance under these alternative measurements. We observe *cosine distance* achieves the best performance. We ascribe this phenomenon to (i) Compared with other alternatives, the *cosine distance* is naturally normalized and bounded, accordingly promising a more stable training; (ii) Mathematically, the *cosine distance* could capture the relative similarity between two vectors in the high dimensional feature space, while other measurements focus on the exact value discrepancy (Huang et al., 2022). Given the inherent capacity gap between the teacher and student, the *cosine distance* is more resilient to feature outliers and serves as a better measurement for our KD framework.

Table 7

Performance of proposed method under the self-KD strategy.

Backbone	Self-KD	FCOS	GFocal	ATSS
ResNet50	✓	66.86 67.66	66.20 67.27	67.40 68.18
ResNet34	✓	66.33 67.14	65.65 67.32	66.99 67.59
ResNet18	✓	61.98 63.60	57.44 63.69	62.57 64.36

4.3.6. Impact of adaptation layer

The adaptation layer ϕ_{ada} is introduced to match the number of feature channels between the student and teacher. This paper considers four ϕ_{ada} types, including 1×1 conv, 1×1 conv+bn, $3 \times 3, 3 \times 3$ +bn. ‘conv’ is the convolutional layer, and ‘+bn’ means that batch normalization is added after the convolutional layer. As the results listed in Table 6(b), the 1×1 conv yields the best student. We surmise the reason is that 1×1 conv could fulfill the recent finding, *consistent teaching* (Beyer et al., 2022), i.e., the student and the teacher having the same spatial context is crucial to reaching peak distillation performance. While 3×3 conv, mixing the spatial information within a 3×3 conv patch renders inconsistency and eventually produces a slightly inferior student. Putting this teaching recipe aside, the worst result is only 0.3 mAP lower than the best, showing that our method is not much sensitive to the choice of the adaptation layer.

4.3.7. Self-KD

In the classical KD, the pre-trained teacher is much stronger than the student. However, the extra training overhead of the teacher limits its practical usage. In this case, self-KD (Kim et al., 2021) has received increasing popularity, in which the student absorbs knowledge from itself, i.e., the teacher and student share the model architectures. Table 7 lists the results under the self-KD settings. On average, our method obtains 2~3% absolute performance promotion over various students. Notably, the GFocal-ResNet18 gains a considerable 6.25% mAP, which is even comparable with the result by distilling from a large teacher (63.69% v.s. 62.68%). These results show the generalization and robustness of our method.

4.3.8. Different distillation settings

We further validate that our method is flexible to be employed across backbone architectures, e.g., heterogeneous teacher–student pairs, and detection frameworks, e.g., two-stage Faster RCNN (Ren et al., 2015). In Fig. 4(a), it is observed that our method works well on both one-stage and two-stage, both anchor-based and anchor-free detectors. On average, 3.7% AP gains are achieved. The main reason for the performance improvements is that ADMD is independent of any ground-truth prior or manual heuristics on specific network architectures, making it more general to different dense detectors. From Fig. 4(b), we notice a larger student is capable of recovering the teacher’s accuracy, while a smaller student could get a higher absolute mAP gain. As shown in Fig. 4(c), we also find that a better teacher tends

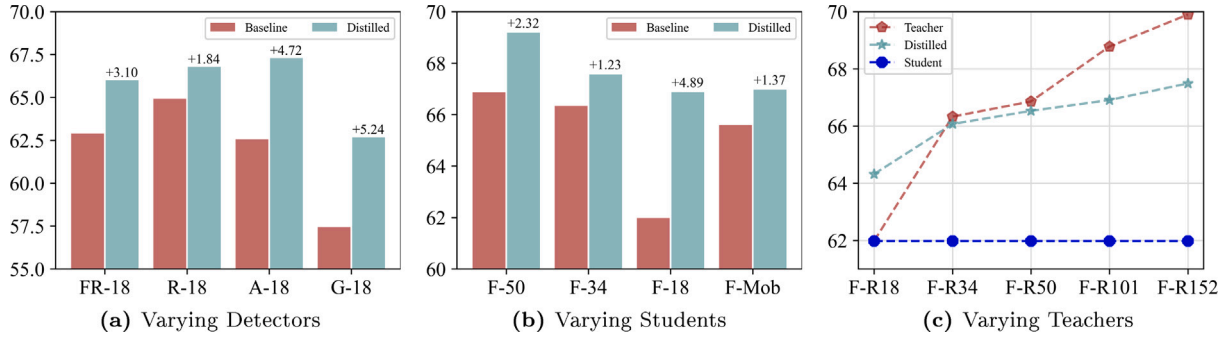


Fig. 4. Results overview of our method by varying the (a) detectors, (b) students, and (c) teachers. The teachers and students share identical detection architecture. The backbone of the teacher is ResNet101 in (a) and (b), and the student is ResNet18 in (c). The X-axis in (a) and (b) represent the students, while in (c) indicates the teacher. The Y-axis denotes the mAP (%). F: FCOS, FR: Faster RCNN, R: RepPoints, A: ATSS, G: GFL, Mob: MobileNet.

Table 8
Experimental results on COCO (Lin et al., 2014).

Method	RetinaNet R101-R50				FCOS R101-R50				GFocal R101-R50			
	mAP	AP _S	AP _M	AP _L	mAP	AP _S	AP _M	AP _L	mAP	AP _S	AP _M	AP _L
Teacher	38.9	21.0	42.8	52.4	40.8	24.2	44.3	52.4	44.9	28.0	49.1	57.2
Student	37.4	20.0	40.7	49.7	38.5	21.9	42.8	48.6	40.2	23.3	44.0	52.2
FitNet	37.4	20.8	40.8	50.9	39.9	23.1	43.4	52.2	40.7	23.7	44.4	53.2
GID	39.1	22.8	43.1	52.3	42.0	25.6	45.8	54.2	41.5	24.3	45.7	53.6
FRS	39.3	21.5	43.3	52.6	40.9	25.7	45.2	51.2	43.6	25.9	47.7	56.4
FGD	39.6	22.9	43.7	53.6	42.1	27.0	46.0	54.6	41.5	24.3	45.7	53.6
MGD	39.5	21.3	43.6	53.4	42.2	26.6	46.3	54.7	42.7	25.2	47.0	55.1
MaskKD	39.8	21.5	43.9	54.0	42.6	26.5	46.9	54.2	40.4	23.5	44.0	52.9
CrossKD	39.7	22.4	43.6	52.8	41.3	25.1	45.5	52.4	43.7	26.9	48.0	56.2
ADMD	40.2	23.1	44.1	53.8	42.0	25.7	46.1	53.3	44.2	27.5	48.7	56.7



Fig. 5. Qualitative performance on DIOR with baseline (top) and the distilled (bottom) FCOS. The undetected and wrongly detected instances of the baseline detector are marked in blue boxes. The teacher is FCOS-ResNet101, and the student is FCOS-ResNet18. Best viewed in color with zoom-in.

to produce better students but is harder to emulate fully. In general, our method consistently accomplishes mAP gains under different scenarios, justifying its versatility and generalization.

4.3.9. Extension on COCO detection

Here, we evaluate our method on the widely used general detection benchmark COCO (Lin et al., 2014) and compare several representative KD approaches devised for COCO detection. As shown in Table 8, our ADMD consistently achieves on par or better distillation results than the sophisticatedly-designed methods in coco detection, accordingly manifesting the efficacy and versatility of our method as a general KD framework that is not confined to RSIs.

4.4. Analysis

4.4.1. Qualitative performance

Fig. 5 compares the qualitative detection performance between the undistilled baseline (top) and a distilled student detector (bottom). Apparently, our method promotes the detectability of the lightweight student model. (i) As shown in the first three columns, the un-distilled model is prone to overlook small objects, while the distilled detector can correctly predict them. We believe this improvement is rooted in that our method could effectively collect the small instances into distillation. (ii) The baseline model fails to locate the instances with ambiguous edges, e.g., the *airport* and *chimney*. With FD for distillation, we alleviate the localization ambiguity and soundly detect these

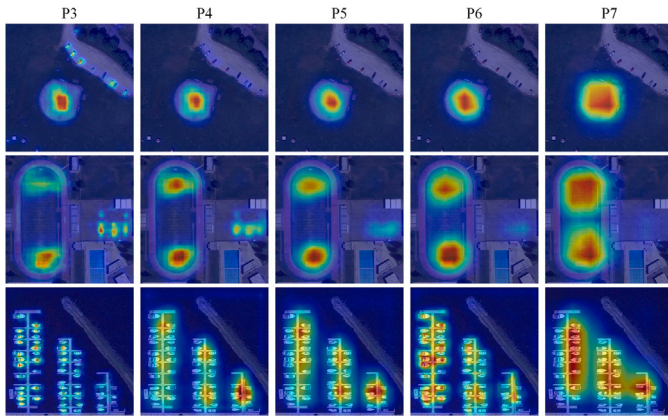


Fig. 6. The corresponding spatial FD in different FPN levels on DIOR. The teacher is FCOS-ResNet101, and the student is FCOS-ResNet18. P3~P7 represents the FPN levels. For clarity, we only visualize the spatial-wise FD within a single iteration, and the exact FD in each iteration is changed during the distillation processes. Best viewed in color with zoom-in.

instances. (iii) In the last column, the student gives a wrong detection in the background area, while our method can avoid this error.

4.4.2. Visualization of spatial-wise FD

To better understand the distillation mask generated in our method, we visualize the heat maps of FD in each FPN level. Fig. 6 shows some examples overlaid on the input images. The properties of FD can be summarized as follows. (i) The FD is a pixel-wise and fine-grained distillation mask, and the key features are emphasized, despite the category and instance size. (ii) As FD does not rely on ground-truth or prior anchors, it can stress some confusing background regions into distillation, which are supposed to be helpful for training detectors (Guo et al., 2021; Yang et al., 2022a; Tian et al., 2019), e.g., reducing false positives. (iii) The instance-related regions, even the small targets, are consistently discovered in the FD. For example, the tiny *vehicle* and *ship* are also gathered, which are typically ignored in previous works (Yang et al., 2022a; Li et al., 2023b). (iv) The FD is clearly distinct in different FPN levels. Specifically, in the low-level FPN layer, the FD focuses on discerning the small instances, e.g., *vehicle*, while the high-level FD concentrates on identifying the large targets, e.g., *baseball court*.

4.4.3. Per-category performance

Fig. 7 visualizes the class-wise AP, showing that the performance gains of the proposed approach are holistic, covering *all* categories. In general, our method is friendly to the difficult categories, e.g., it achieves a sizeable improvement for *TS* (15.6%) and *Dam* (11.2%). One possible reason for this trend is that these hard categories tend to show a large discrepancy between the teacher and the student. Thus, under the guidance of our discrepancy-oriented mask, these categories contribute more to the distillation loss so that the student can better benefit from learning with them. This phenomenon also validates that our method successfully locates profitable pixels and stresses them during KD.

4.4.4. Output discrepancy

One may reasonably wonder if our method really minimizes the output discrepancy between the student and teacher. In short: Yes. We first visualize the object features in Fig. 8. Following the common practice, the features are cropped from the ground-truth areas and then pooled by 2D global averaged pooling. It is observed that the features extracted from our distilled student almost overlap the teacher; namely, the student certainly matches the teacher with our method in the feature space. Besides, the figures indicate that although executing a full

Table 9

Parameter and speed analysis of our method. The FLOPs and FPS are calculated under the input shape of $3 \times 800 \times 800$. #Params denotes the overall parameters of the model, FLOPs is the computational overhead, and FPS represents the evaluation of average frame-per-second on a single NVIDIA GeForce RTX 1080 over 500 trials. ↓: Lower is better, ↑: Higher is better.

Backbone		#Params ↓	FLOPs ↓	FPS ↑	mAP ↑
$\mathcal{T}$	ResNet101	51.25	174.08	16.9	68.78
$\mathcal{S}$	ResNet50	32.26	126.53	22.1	69.18
	ResNet34	29.34	119.14	29.3	67.56
	ResNet18	19.31	95.50	33.9	66.91
	MobileNet	10.64	76.10	28.7	66.97

feature imitation, our method could still aid the student in enlarging the object feature's inter-class separation and intra-class compactness, which directly translates to the elevation of the performance.

We also compute the prediction's variance, both the classification and regression branches, between the student and the teacher for each location. As shown in Fig. 9, one can see that both prior efforts and our method observably reduce the variance. Compared to the previous works, our method helps the student provoke more similar pixel-level predictions to the teacher, thus resulting in the best distilling performance.

4.4.5. Compression performance

Table 9 presents the compression performance. It is observed that notable compression ratio and inference acceleration are achieved with our method; meanwhile, the lightweight detectors are able to recover the teacher's accuracy. For instance, distilled FCOS-MobileNet (Tian et al., 2019) saves 79.2% storage memory and 56.3% computation overhead with slight 2.6% performance drops compared with the teacher. We also mark that the ResNet50-based one somewhat even outstrips the teacher in terms of mAP.

4.4.6. Training efficiency

We mark that our ADMD has a lower training overhead than prior sophisticatedly designed publications. Regarding the KD mask generation, MaskKD (Huang et al., 2023) requires extra training of the mask module, InsDist (Li et al., 2023b) processes each instance in the training images, and TWA (Yang et al., 2024) demands the calculation and storage of gradient information. Besides, most prior methods (Yang et al., 2022d,a; Li et al., 2023b; Ni et al., 2023) include the relation-based KD, which generally involves high-dimensional relational maps build upon each FPN feature. Our ADMD, which is totally training-free and can be implemented within a few lines of code, only needs to measure the feature-level discrepancy and bring no extra module into distillation, thus largely reducing the overall training overhead. In Table 10, we report the memory usage and computation time against several representative detection KD methods. It seems our method enjoys a better trade-off between performance improvement and training efficacy.

5. Conclusion

This article put forth a novel distillation method for learning efficient object detection in remote sensing images. Our method stemmed from the common belief that the difference in features is the reason causing the student to lag behind the teacher. In other words, the student's performance improvements are anticipated when the teacher's and student's features are perfectly aligned, even for the background regions. Therefore, we leveraged the FD to instruct the feature- and logit-level distillation to achieve this goal. Besides, we introduce a simple yet practical weight updating strategy assuring the KD loss term constantly contributes to the student's learning. Experiments conducted on two large-scale datasets attain remarkable performance promotion, showing the superiority of our method over several state-of-the-art

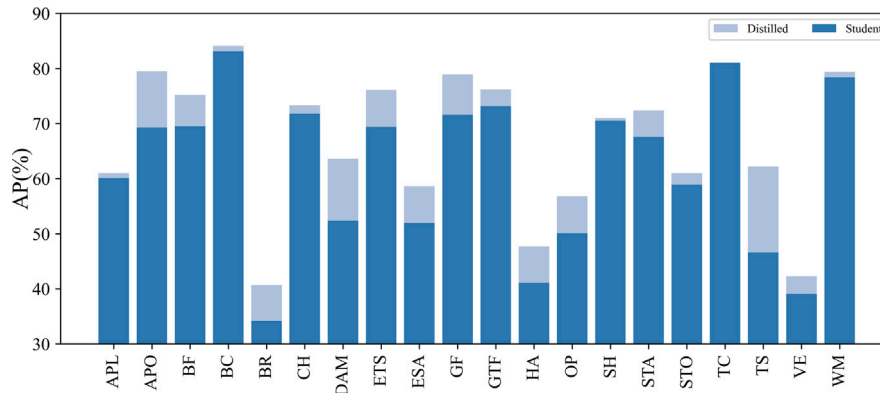


Fig. 7. Per-category performance on DIOR with FCOS ResNet101-ResNet18 as the teacher–student pair. Best viewed in color with zoom-in.

Table 10

Comparison of training efficiency. Teacher and student are ResNet101 and ResNet18 based FCOS (Tian et al., 2019). The GPU memory is calculated under the input shape of $3 \times 800 \times 800$. The training time (per batch) is averaged on a single NVIDIA GeForce RTX 1080 over 500 trials.

Method	KD	FGD	InsDist	MaskKD	DRKD	GKD	TWA	ADMD (Ours)
mAP	63.98	65.88	66.72	66.29	66.03	66.17	66.29	67.05
GPU Memory (G)	7.56	7.21	7.84	9.07	8.84	7.86	8.21	7.23
Training Time (s)	0.85	1.10	1.18	1.09	1.02	1.21	1.16	0.98

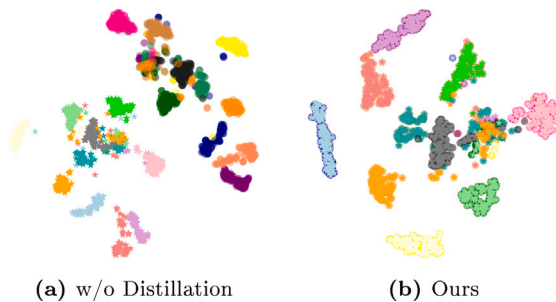


Fig. 8. t-SNE visualization of the embedded features on DIOR. We sample 10 classes. The features of the student and teacher are marked by star (dark color) and dot (light color).

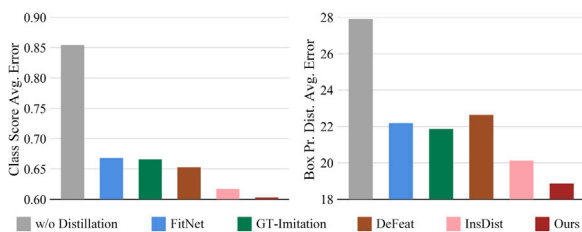


Fig. 9. Per-location averaged L1 errors of classification and bbox scores between the teacher (FCOS-ResNet101) and student (FCOS-ResNet18) on DIOR. Several detection distillation methods are selected for comparison.

detection distillation methods and its versatility across various teacher–student KD pairs. Other possible solutions for the FD measurement and weight adjustment will be investigated in our future works.

CRediT authorship contribution statement

Cong Li: Writing – original draft, Methodology. **Gong Cheng:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Junwei Han:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The author is an Editorial Board Member for ISPRS Journal of Photogrammetry and Remote Sensing and was not involved in the editorial review or the decision to publish this article.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62376223, in part by the Fundamental Research Funds for the Central Universities, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2021B1515020072.

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